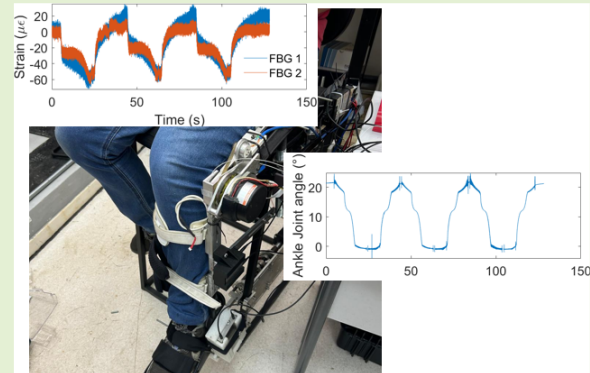


Fiber Bragg Gratings Sensor System for Kinematic and Dynamic Assessment in Lower Limb Robotic Rehabilitation

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Abstract—This article presents the development and application of a fiber Bragg grating (FBG)-based sensor system for dynamic and kinematics assessment of an exoskeleton for lower limb rehabilitation. The exoskeleton has 2 degree-of-freedom, one at the knee joint and the other for the ankle joint, which is positioned on a chair for robotic-assisted rehabilitation of both joints. The proposed FBG-based sensor system includes an FBG array for the strain distribution assessment along the exoskeleton structure, which is used for the human–robot interaction assessment. In addition, FBG accelerometers and inclinometers are positioned on the knee and ankle joints (respectively) of the exoskeleton for the kinematic analysis of such device. The angle characterization tests indicated a root mean squared error (RMSE) for angle assessment of 3.17° and 1.64° for knee and ankle joints, respectively, which results in a relative errors of 3.96% and 6.07% considering the range of movement of both joints. Moreover, the strain sensors indicated a determination coefficient of 0.999 for FBGs 1 and 2. Thereafter, the sensor system is applied on the knee and ankle joint angle measurements as well as the human–robot interaction via strain distribution along the structure. In this case, the tests with a user indicate the variations in knee and ankle joints within the range of movement applied by the exoskeleton’s controller. Furthermore, the interaction forces between the user and the exoskeleton are evaluated using the strain distribution along the exoskeleton structure.

Index Terms—Exoskeleton, fiber Bragg gratings (FBGs), human–robot interaction, kinematics assessment.



I. INTRODUCTION

OPTICAL fiber sensors (OFSs) found numerous applications throughout the years and, especially, in the past

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few years covering different areas of knowledge and implementation, e.g., structural health monitoring [1], industrial instrumentation [2], robotics [3], medicine [4], environmental monitoring [5], just to mention a few. Such popularization of OFS is closely related to the recent widespread of optical components. In addition, the technical advantages of OFS are related to chemical stability, multiplexing capabilities, electromagnetic field immunity, and small dimensions of such approach [6].

Among the different types of OFS, commonly used as interferometers [7], intensity variation sensors [8], and surface plasmon resonance [9] sensors, the distributed and quasi-distributed sensors are used due to the possibility of measuring different parameters and different positions along a single optical fiber cable [10]. Despite the developments in distributed sensors in which optical time-domain [11] and optical frequency-domain reflectometry [12] are commonly used, the fiber Bragg grating (FBG) is a quasi-distributed sensor approach that is one of the most used OFSs due to its additional simplicity on signal acquisition and processing [13].

FBGs are intrinsically sensitive to temperature and strain, and the sensors of such parameters are commonly used using

multiple arrays of FBGs for measuring such parameters in a single optical fiber cable [14]. In addition, different configurations of FBG positioning in structures and devices enable the measurement of all the mechanical parameters, which include transverse strain [15], torsion [16], pressure [17], and forces [18] as well as their use as inclinometers [19], wearable sensing [20] and respiration assessment [21], and accelerometers [22].

Rehabilitation robotics is another important application area for sensors' development, since the robots generally rely on an accurate sensor system as a feedback for robot control [23]. In this case, there is an increase in the reliability of control strategy due to the necessity of direct safe collaboration between human and robot as well as quantitative feedback of patient rehabilitation [24]. For this reason, the robotic rehabilitation systems rely on the kinetic and kinematic feedback from the sensors' system to obtain an accurate control system. Conventionally, potentiometers and encoders are used in position measurement of wearable robots [25]. However, these sensors need to be carefully attached to the robot's joints, since they are sensitive to misalignments, which can result in inaccuracies in sensors' responses [25]. In addition, the additional structures to position potentiometers and encoders can also result in a bulk or less compact system. Inertial measurement units (IMUs) are also used as an alternative for potentiometers and encoders for the estimation of robot's kinematic parameters. However, they need frequent calibration, which is an important drawback in many applications [26]. Furthermore, all the aforementioned sensing technologies have sensitivity to electromagnetic field, which is especially undesirable in robotic applications, where there is a constant activation of electric actuators [27]. The sensitivity to electromagnetic fields is also a drawback for the sensors commonly used on the human-robot interaction forces' assessment (strain gauges, capacitive sensors, piezoelectric sensors) [25]. To that extent, the OFSs have been proposed to overcome some of the drawbacks of electronic and electromechanical sensors with the additional possibility of higher sensitivity and multiplexing data in a single optical fiber cable.

To comply with such performance demands, previous works included efforts on the development of intensity variation-based OFS for microclimate [28], force [27], and angle [29] measurement in exoskeletons and orthosis. Although such sensor systems compare with their electronic counterparts in terms of performance, the use of intensity variation sensors does not include multiplexing capabilities of FBGs and generally presents smaller resolution on the measurand detection. The previous reports on OFSs for exoskeleton instrumentation are mostly based on intensity variation sensors, which present important drawbacks when compared with FBG sensors. To the authors' knowledge, the only report on FBG sensors for exoskeletons is based on polymer optical fibers, which present higher transmission losses and viscoelastic behavior that need additional compensation techniques for signal processing. Moreover, the previous reports of FBG sensors in exoskeleton were based only on force measurement. In addition, FBG sensors present wavelength encoded data, which mitigate the errors on detection due to optical power variations in the light source.

Considering this scenario, FBG sensors in exoskeletons were proposed for single parameter measurement [30]. However, despite initial reports on OFS application in wearable robots, they still lack on integration and multiparameter assessment using a single system. Taking into account this background, this article presents the development of an FBG-based sensor system for full instrumentation of an exoskeleton with 2 degree-of-freedom. To the authors' best knowledge, it is the first report of integrated FBG sensor systems for exoskeleton instrumentation. In this case, the proposed sensor system includes an FBG array for strain distribution assessment along the exoskeleton structure, which is used for human-robot interaction assessment. In addition, FBG accelerometers and inclinometers are positioned on knee and ankle joints of the exoskeleton for kinematic analysis of such device. To that extent, the contributions of this work include a novel sensor system for dynamic and kinematic analysis in exoskeleton using a sensor system with different FBGs, which can be integrated not only for robot control but also for the assessment of patient physical condition through interaction forces between the user and the robot.

II. MATERIALS AND METHODS

The exoskeleton used in the tests is the advanced lower limb orthosis for rehabilitation (ALLOR) positioned on a chair [31], where the user seats on the chair and positions the robotic device on the left knee joint. The orthosis has a closed-loop position control for flexion and extension movements of the knee joint. In addition, there is another joint for ankle movement, which includes a linear series elastic actuator (as shown in [32]) connected to the ankle joint, where the controlled linear movement of the actuator is converted into flexion and extension angles at the ankle joint. Fig. 1 shows a general overview of the proposed exoskeleton, where it is also possible to observe the position of the sensors.

In this case, FBG sensors are positioned in the exoskeleton's structure to measure the human-robot interaction forces and the angles of the knee and ankle joints. Thus, the sensor system comprises an FBG inclinometer positioned on the ankle joint to measure the angles of such joint in the flexion/extension movement. In addition, an FBG accelerometer is positioned on the knee movement, since it has a larger degree of freedom (90° for the knee joint and around 15° for the ankle joint).

A numerical analysis using finite element method (FEM) is performed to obtain the correct positioning of two FBGs in the exoskeleton's structure for human-robot interaction monitoring. The FEM is performed by means of a 1-kN force transversely applied on the exoskeleton structure with a fixed part on the top [see Fig. 2]. In addition, a mesh with 39 056 nodes and 20 148 elements was created considering aluminum as the structure's material. The analysis is performed as a function of equivalent elastic strain distribution along the structure of the exoskeleton. It is important to mention that all the simulations were performed in a commercially available software, Ansys Workbench 2024, which already provides validated FEM analysis and is used only as a preliminary analysis to aid the positioning of the FBG on the structure.

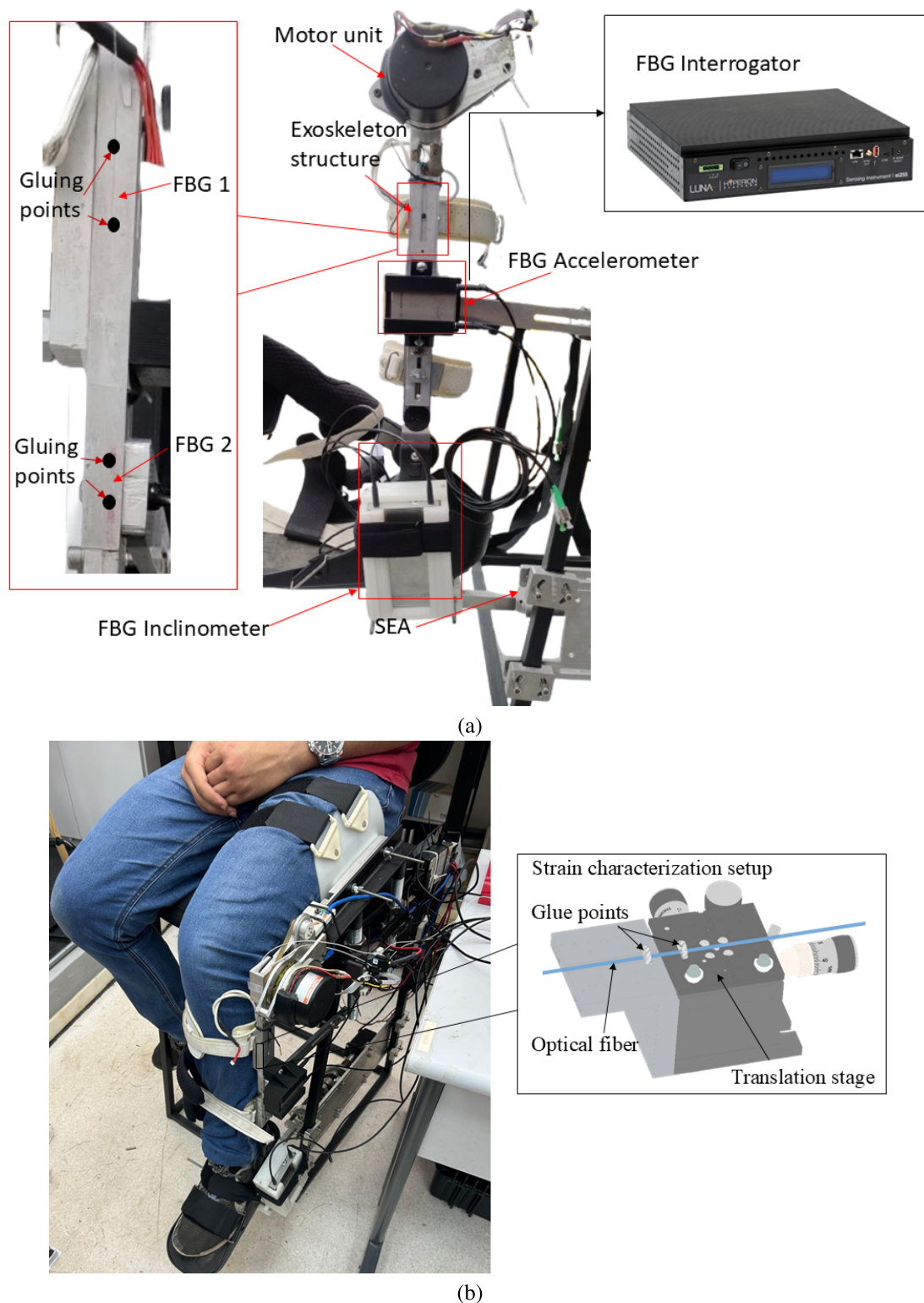


Fig. 1. (a) Exoskeleton overview and sensor positioning. (b) Overview of the user positioned on the exoskeleton with the FBG sensor system. Inset shows the schematic representation of the setup for strain characterization of the FBGs.

The boundary conditions include a fixed support on the top of the structure, which is a condition similar to the final application, where the upper part of the exoskeleton is attached to the chair. In this case, the mesh is created according to an automated mesh generator of the software in which the acceptability of the mesh is assessed by the Jacobian ratio. In addition, the choice of a 1-kN force for the simulation is related to the operational limits of the exoskeleton, which is

not used for a person with weight greater than 1 kN. Although we used the exoskeleton attached to a chair and the 1-kN force is not directly transmitted to the structure, we used it as a parameter for simulation, since it is only a preliminary analysis for the strain that would be applied on the FBG. The criteria for FBG array positioning are based on strain distribution along the structure with the positioning based on the regions with the highest strain on the structure.

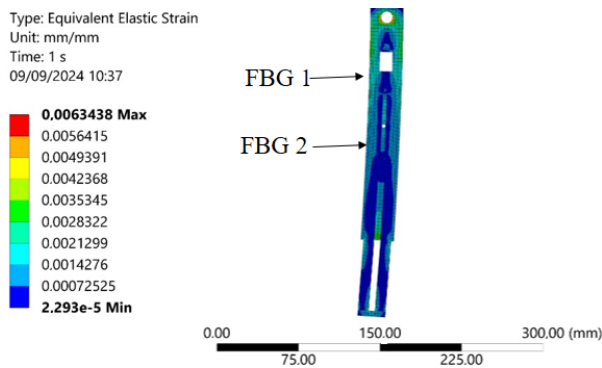


Fig. 2. FEM analysis of strain distribution in the exoskeleton structure under transversal forces.

The setup for FBG inscription of the gratings embedded in the exoskeleton structure comprises a Nd:YAG laser (CNI Laser DPS-266-Q) with an operating wavelength of 266 nm and nanosecond pulses of 8 ns with a repetition rate of 100 Hz and a current of 120 mA. The focal lens (LJ4862-UV, Thorlabs) is used to focus the laser beam to the phase mask (positioned on the focal distance of the lens using two 3-D translation stages (MBT602/M, Thorlabs) and the phase masks are optimized to 266-nm illumination with accuracy and uniformity of 0.01 nm with a period ranging from 1030 to 1060 nm. In addition, three mirrors (NB07-K04, Thorlabs) are used to redirect the laser beam to the focal lens. Such inscription setup was used to inscribe two FBGs (at 1540- and 1568-nm Bragg wavelength), which are positioned on the exoskeleton structure as shown in Fig. 1. The FBGs were attached on the exoskeleton structure using a cyanoacrylate glue on the predefined points for each FBG. The strain sensitivity of both the FBGs is obtained by positioning them on an axial translation stage, where controlled strains are applied on each FBGs (from 0 to 1000 $\mu\epsilon$ in 200- $\mu\epsilon$ steps) and the wavelength shift is analyzed as a function of the applied axial strain. The schematic representation of the setup of strain characterization is presented in Fig. 1 inset, where the fiber is glued in two regions of the translated stage.

Regarding the accelerometer positioned on the knee joint, a commercial FBG accelerometer (os7230, Luna Inc., USA) was used, which also features a temperature compensation using two FBGs in the structure, where temperature cross-sensitivity is mitigated using the direct difference method [33]. A 3D-printed support was developed to provide a repeatable and stable positioning (as also presented in Fig. 1). For angle characterization of the device, static tests are performed in which the knee joint is positioned in predefined angles (from 0° to 80° in 10° steps) and the wavelength shift of the FBG accelerometer is compared with the angles measured by an external goniometer. In addition, data acquisition is performed using a commercial FBG interrogator (Hyperion si255, Luna Inc., USA).

Similarly, the FBG inclinometer (os8100, Luna Inc., USA) with a temperature compensation also using two FBGs and direct difference approach. The sensor characterization as a function of the ankle joint angles was also performed by means of positioning such joint in predefined angles (from 0° to 27° in 3° steps, in this case), where the polynomial regression between the FBG's wavelength shift and the angle (measured

with an external goniometer) was performed via the least-squares method. It is also important to mention that range of the tests is larger than the maximum range of the ankle actuation system to ensure that the sensor is tested for the whole range of application tests.

After characterization of the sensor system, the experimental validation of the proposed device is performed by means of positioning a subject (male, 19 years, 75 kg and 1.70-m height) on the device for knee and ankle flexion and extension movements. It is worth noting that we have the informed consent of the volunteer for tests, which were performed following the guidelines of the National Health Council with the protocols approved by Research Ethics Committee through the National Commission in Research Ethics—CONEP (Certificate of Presentation for Ethical Appreciation—CAAE: 64797816.7.0000.5542). In the validation experiments, sequential cycles are performed in knee and ankle joints. For the knee joint tests, the FBG accelerometer (for knee angle estimation) and the FBG human–robot interaction sensors are analyzed. In contrast, in the ankle joint validation tests, the FBG inclinometer (for ankle angle estimation) is analyzed in conjunction with the FBG human–robot interaction sensors. The sensor system is positioned on the exoskeleton as shown in Fig. 1 and the tests are performed with the user as presented in Fig. 1(b). The tests include two protocols, the first one are the flexion and extension movements of the knee and the second one are flexion and extension movements of the ankle.

III. RESULTS AND DISCUSSIONS

As the first step on the sensor system development, the results of the numerical analysis of strain distribution along the exoskeleton structure are presented in Fig. 2. In this case, a transverse force of 1 kN is applied on the exoskeleton close to the knee joint region to emulate the forces of the user during the exoskeleton tests. Such force resulted in a maximum 634.38 $\mu\epsilon$ in the exoskeleton structure. Furthermore, the stress/strain distribution was considered in the selection of FBG positioning, where the positions FBG1 and FBG2 for the human–robot interaction forces (see Fig. 2) were chosen so are the regions of higher strain of each segment of the exoskeleton's structure. It is important to mention that the chosen force and simulation is based on the maximum force that the exoskeleton can withstand and strain distribution is important not only to estimating a proper positioning for the FBGs but also to verify whether the maximum strain on the structure is larger than the one that FBG can withstand (around 1% strain limit for silica fibers). However, this simulation is the preliminary test for FBG positioning and the validation experiments are focused on the sensor operation instead of FEM simulation validation.

The results of strain characterization in the FBGs used as human–robot interaction sensors are presented in Fig. 3, where it is possible to observe a linear behavior of both the FBGs. In addition, the sensors presented a high repeatability (standard deviation below 2% on the three performed tests demonstrating high repeatability) and the well-known strain sensitivity of 1.1 pm/ $\mu\epsilon$. Furthermore, the root mean squared error (RMSE) is around 8.49 $\mu\epsilon$ and hysteresis of around 0.4% was obtained. After characterization, both the FBGs were

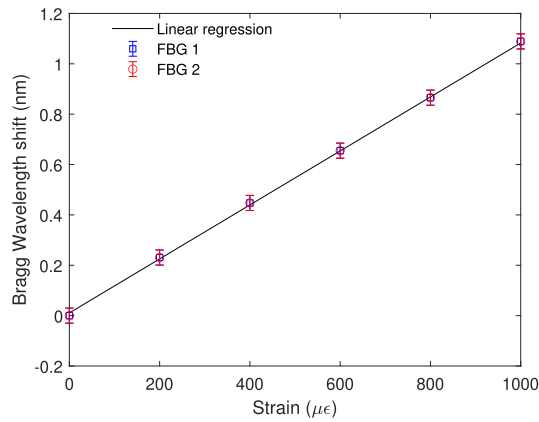


Fig. 3. Bragg wavelength shift as a function of the applied axial strain for both the FBGs used in the human-robot interaction sensor system.

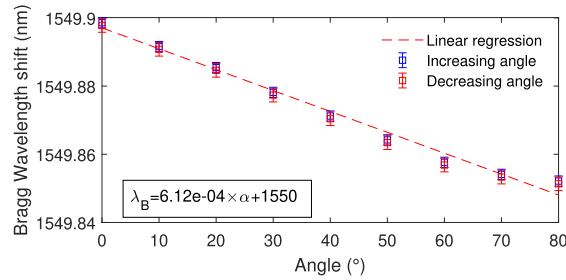


Fig. 4. Wavelength shift as a function of the applied angle on the knee joint of the proposed exoskeleton.

attached to exoskeleton structures in the positions presented in Fig. 2 for the strain assessment as a function of human-robot interaction. In this case, the loading (and forces) applied by the user on the exoskeleton (and vice versa) results in strain distributions on the exoskeleton structure, which are measured by the FBGs.

The wavelength shift of FBG accelerometer as a function of the angle is presented in Fig. 4. The results also presented a linear relationship between the wavelength shift and the angle, where the determination coefficient (R^2) is around 0.992, whereas the sensitivity of $-0.61 \text{ pm}/^\circ$ is obtained. It is also worth to mention that all the tests were performed at constant temperature conditions (around 24°C , room temperature). The results also indicated an accuracy of 3° with 1% hysteresis and standard deviation of around 2.5%.

Then, the FBG inclinometer response is analyzed for different ankle joint angle conditions (at constant temperature condition) as shown in Fig. 5. The sensor analysis indicated a polynomial relationship between both the parameters with R^2 of 0.963. In addition, the error in angle estimation is around 1.64° , which represents a relative error of 6.07%. Moreover, the system presents a small hysteresis of around 0.7% with a standard deviation of 2.3%. In this case, it is possible to observe a nonlinearity on sensor response, especially for angles higher than 20° . The sensor is based on the deformation of the optical fiber as a function of the inclination of a proof mass, and such nonlinearity is related to the saturation of the sensor response, which occurs when the maximum displacement of the proof mass is reached. However, such saturation occurs in angles higher than 20° , which are larger than the ones of the ankle joint for exoskeleton application.

Such saturation can be reduced by changing the proof mass,

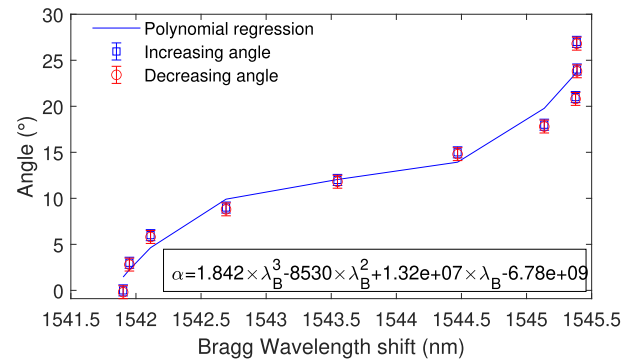
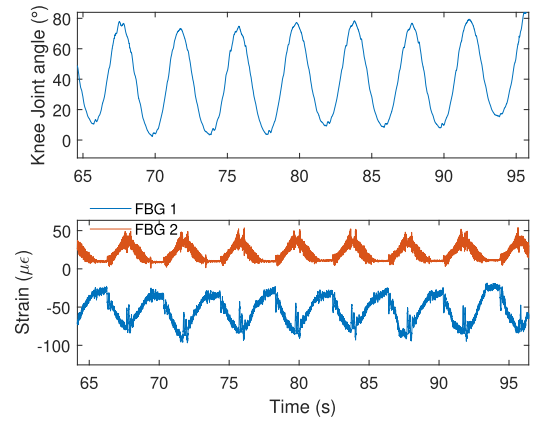
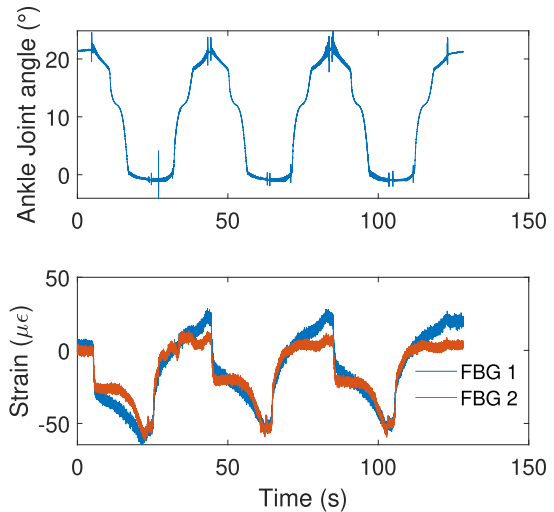


Fig. 5. Wavelength shift of FBG inclinometer as a function of the applied angle on the ankle joint of the proposed exoskeleton.



(a)



(b)

Fig. 6. (a) Results of angle measurement (in knee joint) and human-robot interaction for protocol 1. (b) Results of angle measurement (in ankle joint) and human-robot interaction for protocol 2.

where smaller mass can provide a larger range of movement, but at the cost of smaller sensitivity for angle assessment. Since the maximum variation in the exoskeleton ankle joint is around 20° , the larger proof mass was used to obtain a higher sensitivity within the angle range of the ankle joint.

The results for the tests of each protocol are presented in Fig. 6(a) and (b), respectively. In addition, the results present the sensor responses at different testing days, which indicate the repeatability of the proposed device even in cross-day tests. The results presented in Fig. 6(a) indicate the variation

in the knee angle from 0° to 80° , which is within the range of exoskeleton joint movement. In addition, such angle result is in accordance with the input signals applied to the motor unit. It is worth noting that some of the cycles do not presented a return to the initial angle. However, such behavior can be due to nonlinearity not only on the sensor responses but also on the exoskeleton itself, where inaccuracies in the control system can lead to minor deviation on the angle. The differences in the angles can also be related to the backlash on the exoskeleton structures (especially the gears on the knee joint). Regarding human–robot interaction, both the FBGs in the exoskeleton structure detected the strain in the structure, which is caused by the applied forces from the user to the exoskeleton (and vice versa). The results indicated an extension and compression of the exoskeleton structure under the knee flexion and extension movement and the strain amplitude is related to the applied stress (and forces) to the exoskeleton, indicating a higher human–robot interaction force. It is important to mention that the analysis was made as a function of the strain, since it is the parameter that presents intrinsic sensitivity with the FBG. However, the strain can be converted into forces by means of applying the elastic theory, or by different characterization methods as a function of the force. Such methods are similar to the ones currently used in electronic strain gauges, which are also intrinsic sensitive to strain variations. Furthermore, since both the FBGs are positioned on the structure, it is possible to observe the strain distribution along the exoskeleton when the knee joint movements are performed. In this case, there is a higher strain in FBG 1, i.e., the one closest to the knee joint and FBG 2 follows the same strain pattern, but with smaller amplitude.

Similarly, the angle variation in the ankle joint presented in Fig. 6(b) is also within the range of movement applied on the test (from 0° to 21°). It is important to mention that all the angles and strain measurements are estimated from FBG sensors using the characterizations performed and presented in Figs. 3–5, where the calibration equations presented in the figures are used for angle and strain estimation. However, it is possible to verify some peaks in angle estimation when it reaches values close to 25° and 0° , which are the lower and upper limits for the sensor responses, i.e., the sensor response reaches saturation in values close to the upper and lower limits. Such behavior can be verified in Fig. 5 for angles close to such limits. Comparing the strain results of both the protocols, it is possible to observe a smaller strain on the exoskeleton structure on the ankle joint tests when compared with the ones obtained in the knee joint tests. In contrast, the ankle joint tests indicated similar pattern of both the FBGs with a higher strain in FBG2 when compared with the knee angle variation tests, which is related to the smaller distance of FBG2 to the ankle joint. It is important to mention that the differences in the signal directions comparing Fig. 6(a) and (b), where they are in phase in Fig. 6(b), are related to the direction of the applied forces on the exoskeleton when there are knee and ankle movements. In the knee movements, there is a compression on the FBG closer to the knee joint, whereas the other in the ankle joint has a minor variation related to the extension. In contrast, when the ankle joint flexion and extension movements are performed, the strain of both the FBGs is similar to each other

TABLE I
COMPARISON BETWEEN OFSS' APPROACHES ON
EXOSKELETON'S INSTRUMENTATION

Work	Measurement principle	Multiplexing Capabilities	Number of joints
[28]	Intensity Variation	No	1
[29]	Intensity Variation	No	1
[27]	FBG	Yes	1
This work	FBG	Yes	2

due to the longer distance between the FBGs and the region of the applied force.

The results of the proposed sensor system indicated the feasibility of using FBG arrays for multiparameter monitoring in exoskeletons. The comparison to the previously published sensor systems for similar structures for angle (considering the knee measurement system, since previous works in similar structures did not assess the ankle joint) measurement shows that the proposed device presented accuracy comparable with the potentiometer and intensity variation sensor presented in [27]. In this case, a mean RMSE of 3.7° was found for the previously published devices, whereas the FBG-based sensor presented an RMSE of 3.17° . It is also worth noting that the potentiometer required an additional mechanical structure for the mechanical assembly of the device and connection with the rotation joint and the intensity variation sensor needed additional signal processing and material characterization to compensate viscoelastic effects of the polymer optical fiber, whereas no additional structure, signal processing algorithms, or additional material characterization was needed to translate the wavelength shift into angle variation. Regarding the comparison for strain/force assessment, the results presented in [28] demonstrated a relative error of around 5% for force assessment, which is larger than the one obtained in strain assessment using FBG array (around 1% considering the characterization tests) with the additional advantage of multiplexed strain sensing along the structure of the exoskeleton due to the multiplexing capabilities of the FBGs. The advantages of the proposed approach compared with other types of sensors are summarized in Table I.

IV. CONCLUSION

This article presented the development of an FBG-based sensor system for exoskeleton instrumentation. The exoskeleton is placed on a chair for knee and ankle rehabilitation movements, where the knee and ankle joints are controlled by a motor unit (with harmonic drive) and an SEA, respectively. The sensor system includes the measurement of the angle of the knee and ankle joints as well as the human–robot interaction by means of strain measurement using an array of FBGs in the exoskeleton structure. The sensor system presented an RMSE for angle assessment of 3.17° and 1.64° for knee and ankle joints, respectively, which results in a relative error of 3.96% and 6.07% considering the range of movement of both the joints. Moreover, the strain sensors indicated a determination coefficient of 0.999 for FBGs 1 and 2. Then, the application tests demonstrated the feasibility of the proposed sensor system, which can be readily used on exoskeleton instrumentation with advantages related to electromagnetic immunity (especially important in exoskeletons with electric motors), where such an approach can extent the functionalities of conventional electromechanical sensors

for robots' control. Future works include the implementation of impedance controllers using the FBG sensor systems as feedback and the application in lower limb rehabilitation exercises.

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